

# A fast heuristic approach for train timetabling in a railway node

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## Abstract

We consider a conflict-free scheduling problem which arises in railway networks, where ideal timetables have been provided for a set of trains, but where these timetables may be conflicting. We use a space-time graph approach from the railway scheduling literature in order to develop a fast heuristic which resolves conflicts by adjusting the ideal timetables while attempting to minimize the deviation from the ideal timetable. Our approach is tested on realistic data obtained from the railway node of Milan.

**Keywords:** Train timetabling, Railway network, Heuristic Algorithm.

## 1 Introduction

In this paper we consider the problem of conflict resolution inside a railway node, which is characterized by a number of stations in an urban area interconnected by tracks to form a railway network. An ideal timetable is provided for each train, which specifies the stations to be visited in the railway network together with ideal arrival and departure times at each of the stations. The problem arises when these timetables are conflicting, either due to, for instance, limited station capacity or violation of minimum headway times. The ideal timetables of some trains should therefore be adjusted or some trains canceled in order to reach a feasible timetable which is as close as possible to the ideal timetable.

Cacchiani and Toth [4], very recently, provided an extensive overview of railway optimization problems together with discussions on general solutions approaches and types of problems considered. ILP formulations of similar railway scheduling problems are given by Brännlund *et al.* [1] and Carey and Lockwood [6], while Flier *et al.* [7] considers a problem related to the method used in this paper where an additional train line is included into an existing timetable

on a corridor. As a solution approach we extend the space-time graph approach adopted in Cacchiani *et al.* [3] and in Caprara *et al.* [5].

The contribution of this paper lies in developing a fast heuristic that can solve real-world problem instances within a short period of time. Furthermore, for the first time we take into account specific operational constraints arising from the presence of junctions in the network, as well as the fact that trains are able to travel in both directions between any two stations or use parallel tracks. These are important factors that are usually simplified or omitted in other approaches.

## 2 Problem description

We assume to be given a set of trains  $T$ , a set of stations  $S$ , a set of junctions  $J$  and a set of tracks  $R$  each of which connects two locations<sup>1</sup>, where some of the tracks are parallel tracks (two tracks connecting the same two locations in the same direction). The ideal timetable of a train  $t$  is specified by a set  $L_t \subseteq S \cup J$  of locations to be visited, the order in which they should be visited, the track to be used in case of parallel tracks between stations, an ideal arrival time  $\alpha(t, \ell)$  and an ideal departure time  $\epsilon(t, \ell)$  at each location  $\ell \in L_t$ . Furthermore, for each train  $t$  an ideal profit  $\pi_t$  is given, which represents the profit associated with the train if its timetable is not changed.

The constraints to be satisfied are the following. For the arrival of any two trains  $t, t' \in T$  at the same station  $s \in S$  from track  $r \in R$ , a minimum headway time between the arrival time of the first train and the arrival time of the second train at  $s$  is given and denoted by  $h^+(s, r)$ , while similarly for the departure of the two trains from  $s$  onto  $r$  a minimum headway time between their departure times is given and denoted by  $h^-(s, r)$ . Furthermore, each station  $s \in S$  has a maximum capacity  $c_s$  of trains simultaneously present, while a junction  $j \in J$  has capacity of one with an occupation time of  $o_j$ . Finally, the travel times inside junctions and on tracks connecting locations are assumed to be fixed as specified by the ideal arrival and departure times for each train.

In order to resolve possible conflicts that may occur, trains may be either be *shifted* at their initial locations, *i.e.* depart ahead of schedule or with a delay, trains may be *stretched* at stations they visit, *i.e.* their ideal stopping times at stations may be increased, or trains may be moved from a track onto a parallel track if one exists. Trains are further restricted in terms of shifting and stretching. For each train  $t \in T$  a maximum shift  $s_t^+$  ahead of schedule and maximum shift  $s_t^-$  behind schedule is specified, as well as a maximum total stretch  $\sigma_t$  over all stations. Furthermore, each train is associated with three weights  $\omega_t^+$ ,  $\omega_t^-$  and  $\psi_t$ , which correspond to the penalty for respectively shifting ahead of schedule, shifting behind schedule and stretching.

## 3 Description of the algorithm

The algorithm considers each of the trains individually in a predetermined order (as discussed in the next section). Let  $T^t \subset T$  denote the set of trains that precede train  $t \in T$  in this order.

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<sup>1</sup>In what follows a location will refer to either a station or a junction.

If  $t$  is the  $i$ -th train in this order for  $2 \leq i \leq |T|$ , then during the  $i$ -th iteration of the algorithm an approximate optimal timetable for train  $t$  is determined, subject to minimum headway and capacity constraints imposed by the optimal timetables found during the previous iterations for trains in  $T^t$ . During the first iteration the timetable for the first train is simply its ideal one.

For the purpose of determining an approximate optimal timetable for train  $t \in T$ , we define a *space-time* directed graph  $G = (V, A)$  with node set  $V$  and arc set  $A$ , where each node represents either a departure or an arrival of  $t$  at a specific location from a specific track at a specific point in time, where time is discretized. Each arc of  $G$  either represents  $t$  traveling from one station to another,  $t$  stopping at a station or  $t$  traveling through a junction. For a train  $t$  and a location  $\ell \in L_t$ , let  $\mathcal{A}_{t,\ell}$  denote the set of potentially feasible time instants at which  $t$  may arrive at  $\ell$ , and let  $\mathcal{D}_{t,\ell}$  denote the set of potentially feasible time instants at which  $t$  may depart from  $\ell$ . Hence for  $\tau \in \mathcal{A}_{t,\ell}$  it holds that

$$\alpha(t, \ell) - s_t^+ \leq \tau \leq \alpha(t, \ell) + s_t^- + \sigma_t,$$

while  $\tau \in \mathcal{D}_{t,\ell}$  it holds that

$$\epsilon(t, \ell) - s_t^+ \leq \tau \leq \epsilon(t, \ell) + s_t^- + \sigma_t.$$

For each location  $\ell \in L_t$  such that  $\ell \in S$ , where  $r \in R$  denotes the track used by  $t$  to arrive at  $\ell$ , and each time instant  $\tau \in \mathcal{A}_{t,\ell}$ , there is a node in  $V$  representing the arrival of train  $t$  at  $\ell$  provided that

- (a) there does not exist a train in  $T^t$  which is scheduled to arrive at  $\ell$  on track  $r$  within the time interval  $[\tau - h^+(\ell, r), \tau + h^+(\ell, r)]$ , and
- (b) the number of trains in  $T^t$  which are scheduled to be present at  $\ell$  at time  $\tau$  is less than  $c_\ell$ .

Similarly, for each location  $\ell \in L_t$  such that  $\ell \in S$ , where  $r \in R$  denotes the track used by  $t$  to depart from  $\ell$ , and each time instant  $\tau \in \mathcal{D}_{t,\ell}$ , there is a node in  $V$  representing the departure of train  $t$  from  $\ell$  provided that

- (a) there does not exist a train in  $T^t$  which is scheduled to depart from  $\ell$  on track  $r$  within the time interval  $[\tau - h^-(\ell, r), \tau + h^-(\ell, r)]$ , and
- (b) the number of trains in  $T^t$  which are scheduled to be present at  $\ell$  at time  $\tau$  is less than  $c_\ell$ .

Furthermore, for each location  $\ell \in L_t$  such that  $\ell \in J$  and each time instant  $\tau \in \mathcal{A}_{t,\ell}$ , there is a node in  $V$  representing the arrival of train  $t$  at  $\ell$  provided that there does not exist a train in  $T^t$  which is present at  $\ell$  during the time interval  $[\tau, \tau + o_\ell]$ . Similarly, for each location  $\ell \in L_t$  such that  $\ell \in J$  and each time instant  $\tau \in \mathcal{D}_{t,\ell}$ , there is a node in  $V$  representing the departure of train  $t$  from  $\ell$  provided that there does not exist a train in  $T^t$  which is present at  $\ell$  during the time interval  $[\tau - o_\ell, \tau]$ .

Each arc of  $G$  is associated with a profit, and arcs representing a train traveling between two

locations or through a junction have 0 profit. If, for a location  $l \in L_t$  where  $\ell \in S$ , an arc connects the vertex in  $V$  representing the arrival of train  $t$  at  $l$  at time  $\tau \in \mathcal{A}_{t,\ell}$  to another vertex representing the departure of train  $t$  from location  $l$  at time  $\tau' \in \mathcal{D}_{t,\ell}$ ,  $\tau < \tau'$ , (*i.e.* the arc represents  $t$  stopping at location  $\ell$  from  $\tau$  to  $\tau'$ ), then the associated profit of this arc is equal to

$$-\psi_t(\tau' - \tau - (\epsilon(t, \ell) - \alpha(t, \ell))),$$

*i.e.* the weighted negative value of the stretch at  $\ell$ .

Finally,  $V$  contains a dummy starting vertex  $v_s$  and a dummy ending vertex  $v_e$ . If  $\ell \in L_t$  is the initial location of train  $t$ , then for each  $\tau \in \mathcal{D}_{t,\ell}$  there exists an arc connecting  $v_s$  to the vertex representing the departure of  $t$  from  $\ell$  at time  $\tau$ . The profit of this arc is equal to  $\pi_t - \omega_t^-(\tau - \epsilon(t, \ell))$  if  $\tau \geq \epsilon(t, \ell)$ , and  $\pi_t - \omega_t^+(\epsilon(t, \ell) - \tau)$  otherwise, *i.e.* the weighted shift of  $t$  at  $\ell$  subtracted from the ideal profit of  $t$ . If  $\ell \in L_t$  is the final location of train  $t$ , then for each  $\tau \in \mathcal{A}_{t,\ell}$  there exists an arc connecting the vertex representing the arrival of  $t$  at  $l$  at time  $\tau$  to  $v_e$ , where this arc has an associated profit of 0.

The problem under consideration in this paper is an extension of the problem considered in [5], where the authors addressed the *Train Timetabling Problem* (TTP) for a single corridor imposing just a subset of constraints arising from real world applications. In this paper the authors proved that the TTP is NP-hard and they introduced the following mathematical formulation of the problem (reported here for sake of completeness) using a binary variable  $x_a$  which takes a value of 1 if a train  $t$  is using arc  $a \in A$ . Moreover  $\delta_t^+(v)$  denotes the set of feasible leaving arcs for node  $v \in V$  and  $\delta_t^-(v)$  the set of feasible entering arcs, given a specific train  $t \in T$ .

$$\text{maximize} \quad \sum_{a \in A} p_a x_a \quad (1)$$

$$\text{subject to} \quad \sum_{a \in \delta_t^+(v_s)} x_a \leq 1 \quad t \in T \quad (2)$$

$$\sum_{a \in \delta_t^-(v)} x_a - \sum_{a \in \delta_t^+(v)} x_a = 0 \quad t \in T, v \in V \setminus \{v_s, v_e\} \quad (3)$$

$$\sum_{a \in \tilde{C}} x_a \leq 1 \quad \tilde{C} \in \tilde{\mathcal{C}} \quad (4)$$

$$x_a \in \{0, 1\} \quad a \in A \quad (5)$$

The objective function (1) maximizes the total profit, where  $p_a$  gives the profit of arc  $a \in A$  as described above in this section. Constraints (2) and (3) ensure that if a train is scheduled, it follows a feasible path. Finally *clique constraints* (4) (see [5]) forbid the simultaneous selection of incompatible arcs imposing the set of operational constraints (described in Section 2) — in particular the new operational constraints arising from the dynamics inside junctions.

Due to the fact that this mathematical formulation is characterized by large numbers of constraints

and variables, which prevents the use thereof in real world applications, we decided to adopt a heuristic approach based on the following idea. Given a specific ordering of trains, the algorithm tries to schedule these trains one after the other, each time taking into consideration operational constraints imposed by previously scheduled ones. Given the space-time graph for a train  $t$  during an iteration of the algorithm, an approximate longest (maximum profit) path from  $v_s$  to  $v_e$  in this directed, acyclic graph is heuristically computed using a relaxed dynamic programming approach. If no such path exists, or if the total profit of the path is negative, the train is canceled. Otherwise, the train timetable is updated according to its optimal path.

## 4 Preliminary results

In this section we present some preliminary results obtained by using realistic data provided by RFI (Rete Ferroviaria Italiana, the main railway infrastructure manager in Italy) concerning the railway node of Milan. We consider an instance where there are 53 stations, 10 junctions and 146 tracks (14 parallel tracks) connecting these locations. Moreover, ideal timetables are given for 1 500 trains for a period of one day, where each time instant represents a minute of the day. Also, the maximum shift ahead of schedule is equal to 1 minute, the maximum shift behind schedule is 10 minutes and the maximum stretch is 2 minutes. Finally, for the ideal profits for each train and the penalty weights for shifting and stretching, we use the realistic values used in [2] (shown here in Table 1). The algorithm was implemented in C++ and run in Linux Ubuntu (compiled using g++ with option -O3) on an Intel Core i3-2330M CPU computer clocked at 2.20GHz, with 4 GB of RAM.

| Train type              | Eurostar | Euronight | Intercity | Express | Combined | Direct | Local | Freight |
|-------------------------|----------|-----------|-----------|---------|----------|--------|-------|---------|
| $pi_t$                  | 200      | 150       | 120       | 110     | 100      | 100    | 100   | 100     |
| $\omega_t^+/\omega_t^-$ | 7        | 7         | 6         | 5       | 6        | 5      | 5     | 2       |
| $\psi_t$                | 10       | 10        | 9         | 8       | 9        | 8      | 6     | 3       |

**Table 1:** Profits and penalty weights for each train type

To improve the quality of the solutions obtained, the algorithm is repeated with different examination orders of the trains, henceforth referred to as a *multi-run*. Two different methods for generating random orders were used, as well as a first-come-first-serve (FCFS) order. In one method for generating random orders trains are partitioned into two classes, one of which is considered to be more important than the other (priority ordering). In this case a random permutation of the more important trains are generated and examined first, followed by a random permutation of the second class of trains. In the second method for generating random orders, no distinction between the trains is made and a random permutation of all the trains is generated.

The preliminary results obtained using our algorithm and the three ordering strategies mentioned above (priority, no priority and FCFS) is shown in Table 2 for multi-runs consisting of 10, 100 and 1000 iterations. The table shows the total profit obtained summed over all the trains that were not canceled, and also contains information regarding the total number of trains canceled,

the number of important trains canceled, and the number of trains canceled due to a negative profit.

The total time and the time to obtain the best solution is shown (both expressed in seconds), as well as the number of different solutions obtained by the algorithm. Furthermore, the total stretch over all the trains in the best solution found is given, together with the number of trains for which the ideal timetable was changed as well as the total number of parallel tracks used. Finally in order to give an indication of the quality of the solutions found, it is worth mentioning that a bound on the optimal solution value can easily be computed by summing the profits of all the trains. The percentage gap between this bound and the heuristic solution values obtained is given in the final column of the table.

|             | Profit | Iter. | Trains<br>can. | Imp.<br>trains<br>can. | Can.<br>due to<br>profit | Time<br>to sol. | Diff.<br>sol. | Time<br>total | Total<br>str. | Trains<br>aff. | Tracks<br>changed | Gap    |
|-------------|--------|-------|----------------|------------------------|--------------------------|-----------------|---------------|---------------|---------------|----------------|-------------------|--------|
| FCFS        | 144562 | 1     | 151            | 1                      | 53                       | 0.3             | 1             | 0.3           | 4867          | 567            | 116               | 12.01% |
| No priority | 145676 | 10    | 158            | 5                      | 20                       | 0.6             | 10            | 3.2           | 2934          | 505            | 98                | 11.34% |
|             | 146147 | 100   | 152            | 7                      | 28                       | 14.1            | 100           | 32.9          | 2857          | 511            | 128               | 11.05% |
|             | 146846 | 1000  | 145            | 4                      | 23                       | 135.8           | 817           | 333.1         | 3021          | 537            | 120               | 10.62% |
| Priority    | 145561 | 10    | 160            | 0                      | 26                       | 2.7             | 10            | 3.4           | 2913          | 509            | 115               | 11.41% |
|             | 146383 | 100   | 151            | 0                      | 23                       | 6.8             | 99            | 33.0          | 2953          | 531            | 121               | 10.91% |
|             | 147034 | 1000  | 143            | 0                      | 20                       | 100.7           | 817           | 326.8         | 3217          | 542            | 110               | 10.51% |

**Table 2:** Results obtained for the node of Milan

## 5 Conclusion

In this paper we presented a heuristic for solving the conflict free scheduling problem on a railway network. Results were obtained using realistic data from the node of Milan. The results at this stage are preliminary and future work include an extension of Table 2 as well as extensions in the algorithm. This will include allowing trains the possibility to be rerouted when there are no feasible paths in the space-time graph. For this purpose alternative routes for each train have to be calculated or given as input to the problem. Furthermore, apart from resolving conflicts in given timetables, the algorithm proposed in this paper may also be used as a planning tool for testing network capacities or for real time conflict resolution in case of disruptions on a railway network. In the latter case the approach would especially prove useful due to the fact that the computation times are small.

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